

# Benchmark Comparison Guide: PortX MuleSoft JDE Connector → Atina JDE Connector

## Table of Contents

|                                                |    |
|------------------------------------------------|----|
| 1. Purpose and scope .....                     | 2  |
| 2. Benchmark methodology and environment ..... | 2  |
| 2.1. Test environment .....                    | 2  |
| 2.2. Warm-up and caching behavior .....        | 2  |
| 2.3. How averages are calculated .....         | 3  |
| 3. Scenario 01: Get Address Book .....         | 3  |
| 3.1. APIs used .....                           | 3  |
| 3.2. Cycle 1 .....                             | 4  |
| 3.3. Cycle 2 .....                             | 4  |
| 4. Scenario 02: Create Address Book .....      | 5  |
| 4.1. APIs used .....                           | 5  |
| 4.2. Cycle 1 .....                             | 5  |
| 4.3. Cycle 2 .....                             | 6  |
| 5. Scenario 03: Delete Address Book .....      | 6  |
| 5.1. APIs used .....                           | 6  |
| 5.2. Cycle 1 .....                             | 7  |
| 5.3. Cycle 2 .....                             | 7  |
| 6. Scenario 04: Get Customer .....             | 8  |
| 6.1. APIs used .....                           | 8  |
| 6.2. Cycle 1 .....                             | 8  |
| 6.3. Cycle 2 .....                             | 9  |
| 7. Scenario 05: Create Customer .....          | 9  |
| 7.1. APIs used .....                           | 9  |
| 7.2. Cycle 1 .....                             | 10 |
| 7.3. Cycle 2 .....                             | 10 |
| 8. Scenario 06: Delete Customer .....          | 11 |
| 8.1. APIs used .....                           | 11 |
| 8.2. Cycle 1 .....                             | 11 |
| 9. Scenario 07: Create Sales Order .....       | 12 |
| 9.1. APIs used .....                           | 12 |
| 9.2. Cycle 1 .....                             | 13 |
| 9.3. Cycle 2 .....                             | 13 |

|                                                       |    |
|-------------------------------------------------------|----|
| 10. Scenario 08: Submit UBE and retrieve status ..... | 13 |
| 10.1. Cycle 1 .....                                   | 14 |
| 10.2. Cycle 2 .....                                   | 14 |
| 11. Scenario 09: Get Events (MBF Events) .....        | 14 |
| 11.1. Cycle 1 .....                                   | 15 |
| 11.2. Cycle 2 .....                                   | 15 |
| 12. Summary: Which connector performs better? .....   | 16 |

# 1. Purpose and scope

This document helps customers understand:

- What changes (and what does **not** change) when migrating from the PortX (ModusBox) MuleSoft JDE Connector to the Atina JDE Connector.
- A side-by-side performance comparison under the **same** local test conditions.

## IMPORTANT

The Atina JDE Connector includes an additional integration option not available in the PortX (ModusBox) connector: **JDE Web Services (BSSV)** invocation. The Web Services scenarios are included for comparison and context. In real projects, Web Services are often a good fit for **transaction creation** because they encapsulate business rules similarly to interactive applications (pricing, invoicing, sales order logic, etc.).

# 2. Benchmark methodology and environment

## 2.1. Test environment

- Client machine: Mac (M4)
- Mule runtime: executed from Anypoint Studio
- Atina Microservice: Docker Desktop
- Test runner: Postman

These results are intended to compare connectors under identical conditions.

## IMPORTANT

In a real deployment, the Atina Microservice is typically placed closer (network-wise) to the JDE Enterprise Server / Logic tier, which usually improves overall performance versus this local setup.

## 2.2. Warm-up and caching behavior

The first execution cycle often shows higher latency due to warm-up effects (JIT, connection setup,

metadata caching, etc.). Subsequent cycles are generally more representative of steady-state behavior.

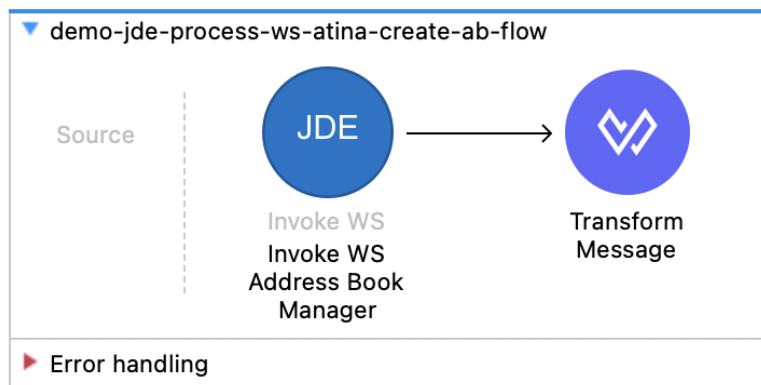
## 2.3. How averages are calculated

- If a run is marked as **Time Out**, it is excluded from the **Average** calculation for that API column.
- If all runs for a given API are Time Out / invalid, the average is shown as **Error**.

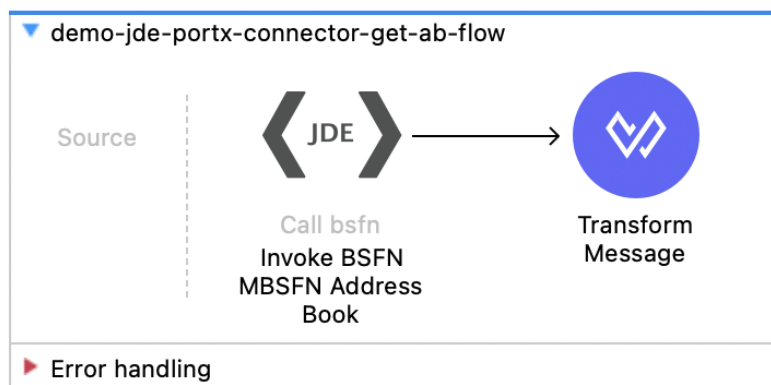
## 3. Scenario 01: Get Address Book

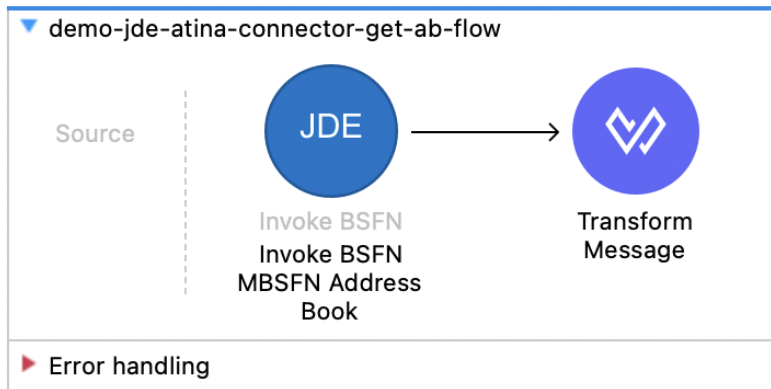
### 3.1. APIs used

- Web Service:
  - `oracle.e1.bssv.JP010000.AddressBookManager.getAddressBook`



- BSFN (MBF):
  - `AddressBookMasterMBF` (Action: **Inquire**) for both Atina and PortX





## 3.2. Cycle 1

| Execution           | Get Address Book Using Web Services (ms) | Get Address Book MBF - Atina (ms) | Get Address Book MBF - PortX (ms) |
|---------------------|------------------------------------------|-----------------------------------|-----------------------------------|
| Exec 1              | 48822                                    | 4811                              | 58955                             |
| Exec 2              | 19062                                    | 434                               | 540                               |
| Exec 3              | 12887                                    | 414                               | 533                               |
| <b>Average (ms)</b> | <b>26923.67</b>                          | <b>1886.33</b>                    | <b>20009.33</b>                   |

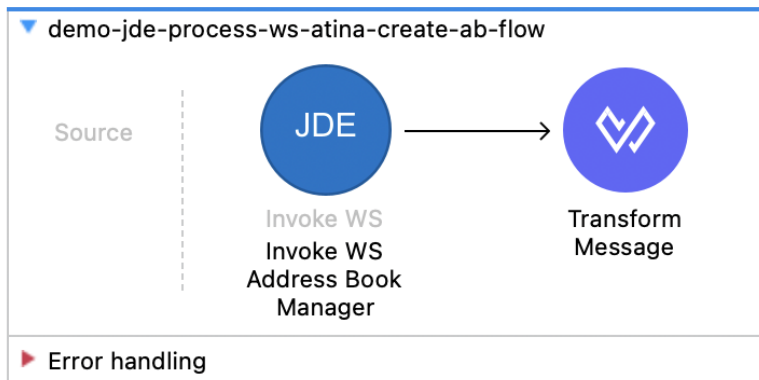
## 3.3. Cycle 2

| Execution           | Get Address Book Using Web Services (ms) | Get Address Book MBF - Atina (ms) | Get Address Book MBF - PortX (ms) |
|---------------------|------------------------------------------|-----------------------------------|-----------------------------------|
| Exec 1              | 13856                                    | 706                               | 638                               |
| Exec 2              | 11291                                    | 444                               | 704                               |
| Exec 3              | 13756                                    | 376                               | 584                               |
| Exec 4              | 13830                                    | 419                               | 640                               |
| Exec 5              | 11059                                    | 401                               | 577                               |
| Exec 6              | 18287                                    | 421                               | 582                               |
| Exec 7              | 25588                                    | 428                               | 4925                              |
| Exec 8              | 11010                                    | 469                               | 610                               |
| Exec 9              | 13050                                    | 417                               | 582                               |
| Exec 10             | 11099                                    | 411                               | 621                               |
| <b>Average (ms)</b> | <b>14282.60</b>                          | <b>449.20</b>                     | <b>1046.30</b>                    |

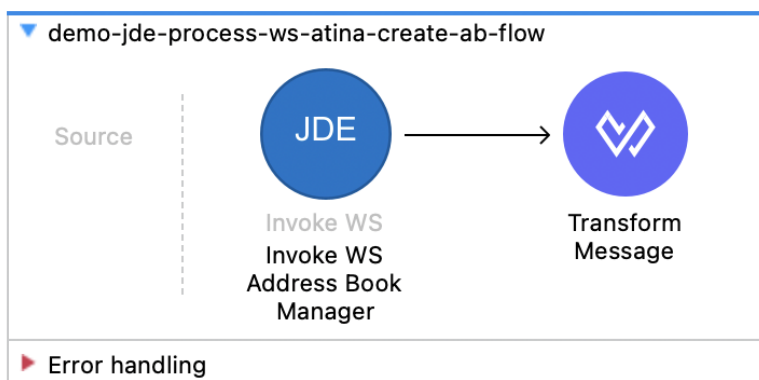
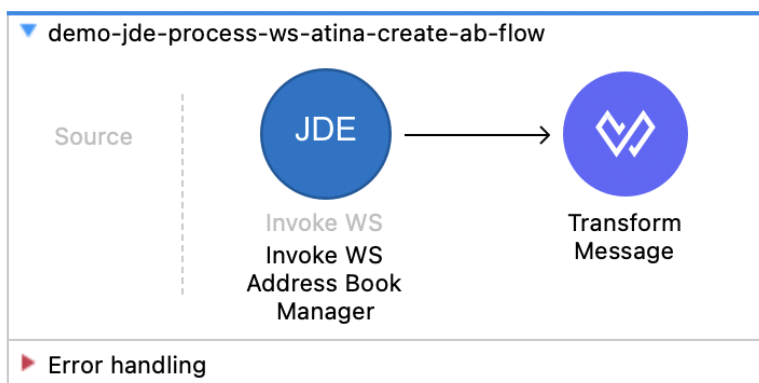
## 4. Scenario 02: Create Address Book

### 4.1. APIs used

- Web Service:
  - `oracle.e1.bssv.JP010020.CustomerManager.processCustomer`



- BSFN (MBF):
  - `AddressBookMasterMBF` (Action: **Add**) for both Atina and PortX



### 4.2. Cycle 1

| Execution    | Create AB WS (ms) | Create AB BSFN - Atina (ms) | Create AB BSFN - PortX (ms) |
|--------------|-------------------|-----------------------------|-----------------------------|
| Exec 1       | 28594             | 1497                        | 12973                       |
| Exec 2       | 5585              | 522                         | 631                         |
| Exec 3       | 5472              | 5152                        | 628                         |
| Average (ms) | 13217.00          | 2390.33                     | 4744.00                     |

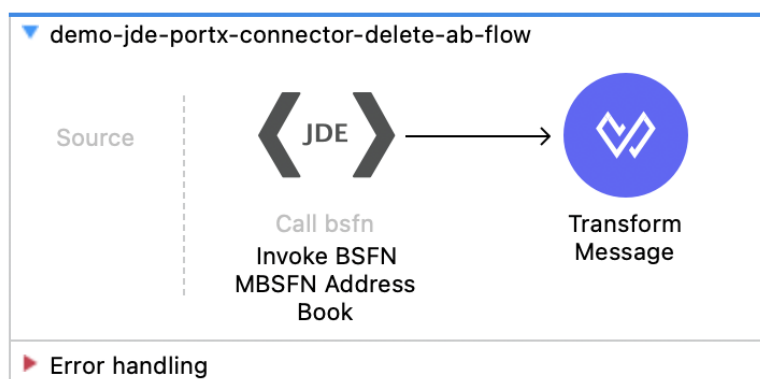
## 4.3. Cycle 2

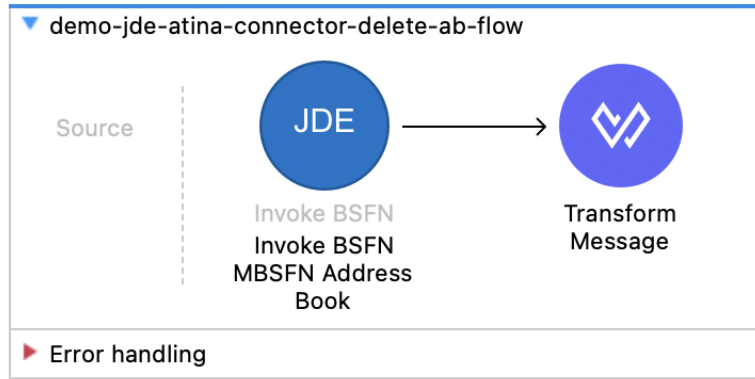
| Execution    | Create AB Using Web Services (ms) | Create AB Using BSFN - Atina (ms) | Create AB Using BSFN - PortX (ms) |
|--------------|-----------------------------------|-----------------------------------|-----------------------------------|
| Exec 1       | 13207                             | 526                               | 4981                              |
| Exec 2       | 5529                              | 501                               | 706                               |
| Exec 3       | 9947                              | 531                               | 639                               |
| Exec 4       | 5403                              | 481                               | 618                               |
| Exec 5       | 5733                              | 498                               | 707                               |
| Exec 6       | 13037                             | 523                               | 5106                              |
| Exec 7       | 5543                              | 481                               | 620                               |
| Exec 8       | 5381                              | 535                               | 614                               |
| Exec 9       | 5475                              | 480                               | 661                               |
| Exec 10      | 5464                              | 456                               | 627                               |
| Average (ms) | 7471.9                            | 501.2                             | 1527.9                            |

# 5. Scenario 03: Delete Address Book

## 5.1. APIs used

- BSFN (MBF):
  - **AddressBookMasterMBF** (Action: **Delete**) for both Atina and PortX





## 5.2. Cycle 1

| Execution           | Delete AB Using BSFN - Atina (ms) | Delete AB Using BSFN - PortX (ms) |
|---------------------|-----------------------------------|-----------------------------------|
| Exec 1              | 1297                              | 1466                              |
| Exec 2              | 656                               | 747                               |
| Exec 3              | 622                               | 1362                              |
| <b>Average (ms)</b> | <b>858.33</b>                     | <b>1191.67</b>                    |

## 5.3. Cycle 2

### NOTE

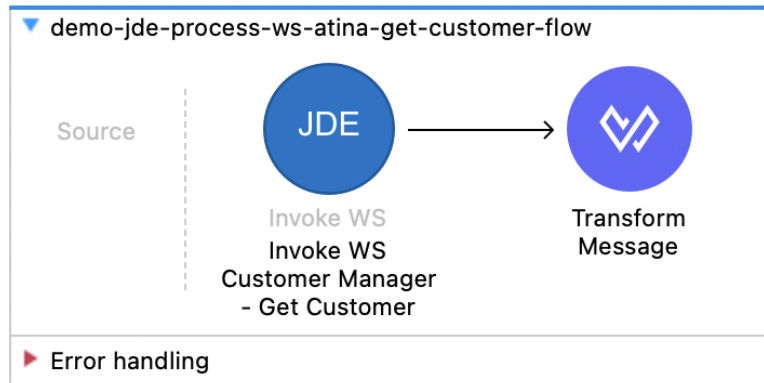
This cycle contains **two** Atina delete runs (different executions/records). They are shown separately.

| Execution           | Delete AB Using BSFN - Atina (Run A) (ms) | Delete AB Using BSFN - Atina (Run B) (ms) | Delete AB Using BSFN - PortX (ms) |
|---------------------|-------------------------------------------|-------------------------------------------|-----------------------------------|
| Exec 1              | 592                                       | 641                                       | 1293                              |
| Exec 2              | 7414                                      | 675                                       | 743                               |
| Exec 3              | 668                                       | 660                                       | 725                               |
| Exec 4              | 635                                       | 617                                       | 726                               |
| Exec 5              | 620                                       | 577                                       | 790                               |
| Exec 6              | 637                                       | 583                                       | 714                               |
| Exec 7              | 631                                       | 625                                       | 727                               |
| Exec 8              | 656                                       | 552                                       | 746                               |
| Exec 9              | 629                                       | 567                                       | 708                               |
| Exec 10             | 579                                       | 586                                       | 753                               |
| <b>Average (ms)</b> | <b>1306.10</b>                            | <b>608.30</b>                             | <b>792.50</b>                     |

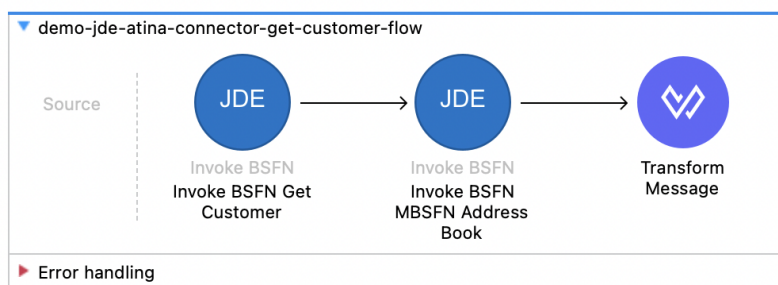
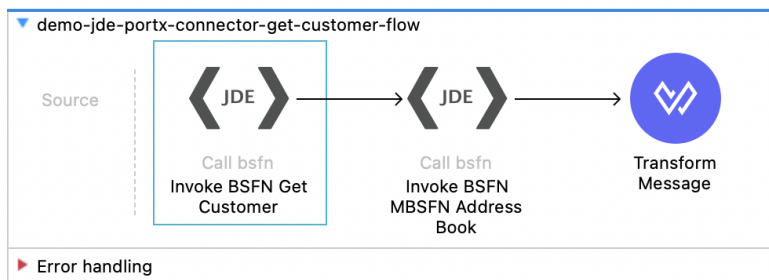
## 6. Scenario 04: Get Customer

### 6.1. APIs used

- Web Service:
  - `oracle.e1.bssv.JP010020.CustomerManager.getCustomerV3`



- BSFNs:
  - `MBFCustomerMaster` + `AddressBookMasterMBF` (Action: **Inquire**) for both Atina and PortX



### 6.2. Cycle 1

| Execution | Get CUSTOMER WS<br>(ms) | Get CUSTOMER BSFN -<br>Atina (ms) | Get CUSTOMER BSFN -<br>PortX (ms) |
|-----------|-------------------------|-----------------------------------|-----------------------------------|
| Exec 1    | 59452                   | 995                               | Time Out                          |
| Exec 2    | 18825                   | 5438                              | 16212                             |
| Exec 3    | 35719                   | 859                               | 5413                              |



| Execution    | Get CUSTOMER WS (ms) | Get CUSTOMER BSFN - Atina (ms) | Get CUSTOMER BSFN - PortX (ms) |
|--------------|----------------------|--------------------------------|--------------------------------|
| Average (ms) | 37998.67             | 2430.67                        | 10812.50                       |

**NOTE** Averages are calculated excluding the execution marked as Time Out.

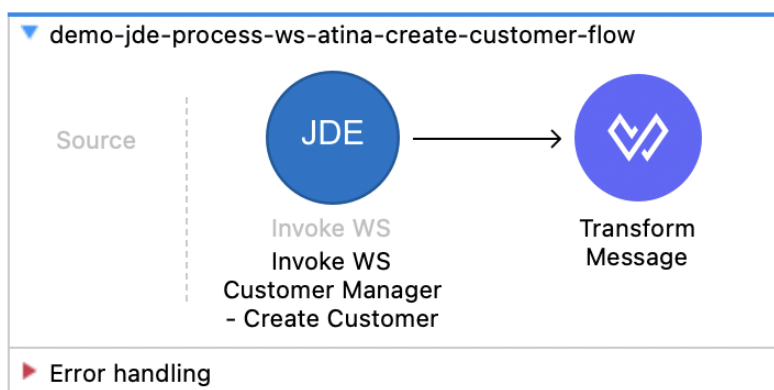
## 6.3. Cycle 2

| Execution    | Get CUSTOMER WS (ms) | Get CUSTOMER BSFN - Atina (ms) | Get CUSTOMER BSFN - PortX (ms) |
|--------------|----------------------|--------------------------------|--------------------------------|
| Exec 1       | 39874                | 866                            | 9941                           |
| Exec 2       | 18538                | 818                            | 1142                           |
| Exec 3       | 16983                | 856                            | 1135                           |
| Exec 4       | 16446                | 849                            | 1118                           |
| Exec 5       | 18420                | 817                            | 1118                           |
| Exec 6       | 17300                | 883                            | 1168                           |
| Exec 7       | 18397                | 859                            | 1125                           |
| Exec 8       | 16480                | 862                            | 1162                           |
| Exec 9       | 18505                | 855                            | 1108                           |
| Exec 10      | 20067                | 858                            | 1162                           |
| Average (ms) | 20101.00             | 852.30                         | 2017.90                        |

## 7. Scenario 05: Create Customer

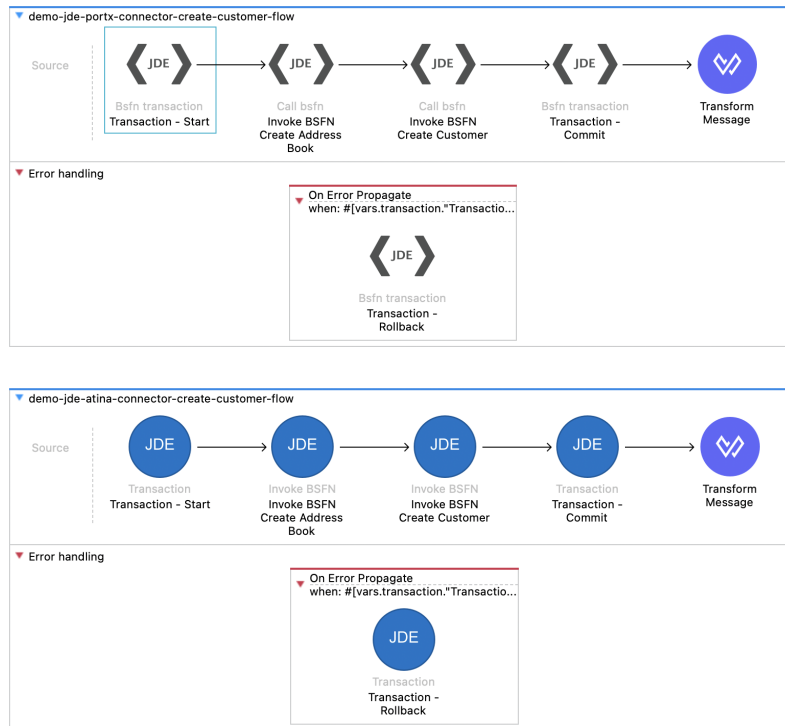
### 7.1. APIs used

- Web Service:
  - `oracle.e1.bssv.JP010020.CustomerManager.processCustomer`



- BSFNs:

- **MBFCustomerMaster** + **AddressBookMasterMBF** (Action: **Add**) for both Atina and PortX



## 7.2. Cycle 1

| Execution           | Create CUSTOMER WS (ms) | Create CUSTOMER BSFN - Atina (ms) | Create CUSTOMER BSFN - PortX (ms) |
|---------------------|-------------------------|-----------------------------------|-----------------------------------|
| Exec 1              | 30484                   | 3445                              | 14928                             |
| Exec 2              | 13346                   | 1956                              | Time Out                          |
| <b>Average (ms)</b> | <b>21915.00</b>         | <b>2700.50</b>                    | <b>14928.00</b>                   |

**NOTE** | Averages are calculated excluding the execution marked as Time Out.

## 7.3. Cycle 2

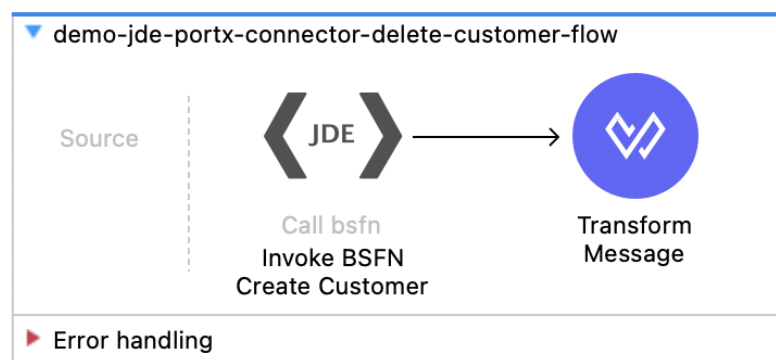
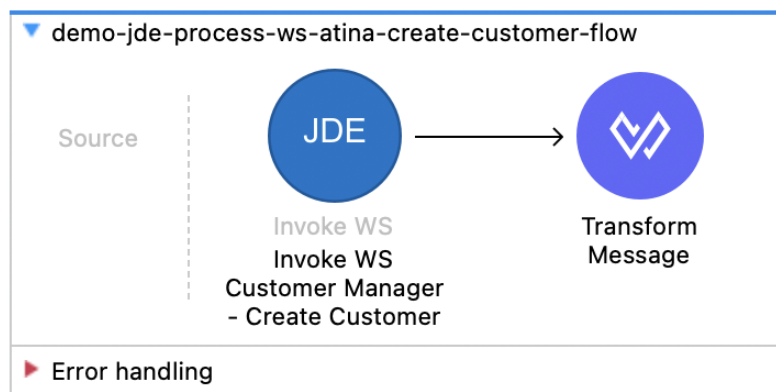
| Execution | Create CUSTOMER WS (ms) | Create CUSTOMER BSFN - Atina (ms) | Create CUSTOMER BSFN - PortX (ms) |
|-----------|-------------------------|-----------------------------------|-----------------------------------|
| Exec 1    | 27399                   | 2020                              | 17058                             |
| Exec 2    | 4980                    | 1946                              | 1464                              |
| Exec 3    | 6837                    | 1986                              | 1405                              |
| Exec 4    | 4999                    | 1943                              | 1440                              |
| Exec 5    | 6905                    | 2022                              | 1435                              |
| Exec 6    | 4976                    | 1929                              | 1468                              |
| Exec 7    | 4882                    | 1963                              | 1355                              |

| Execution    | Create CUSTOMER WS (ms) | Create CUSTOMER BSFN - Atina (ms) | Create CUSTOMER BSFN - PortX (ms) |
|--------------|-------------------------|-----------------------------------|-----------------------------------|
| Exec 8       | 6909                    | 1983                              | 1375                              |
| Exec 9       | 5011                    | 1919                              | 1378                              |
| Exec 10      | 42296                   | 2912                              | 11381                             |
| Average (ms) | 11519.40                | 2062.30                           | 3975.90                           |

## 8. Scenario 06: Delete Customer

### 8.1. APIs used

- BSFNs:
  - MBFCustomerMaster** + **AddressBookMasterMBF** (Action: **Delete**) for both Atina and PortX



### 8.2. Cycle 1

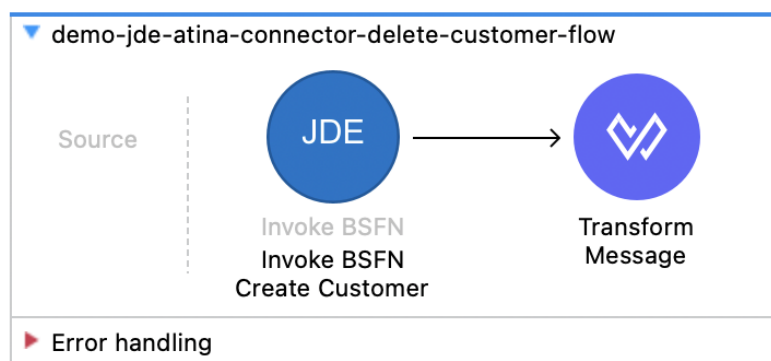
| Execution | Delete CUSTOMER BSFN - Atina (ms) | Delete CUSTOMER BSFN - PortX (ms) |
|-----------|-----------------------------------|-----------------------------------|
| Exec 1    | 698                               | 1373                              |
| Exec 2    | 755                               | 884                               |
| Exec 3    | 742                               | 835                               |
| Exec 4    | 748                               | 898                               |

| Execution           | Delete CUSTOMER BSFN - Atina (ms) | Delete CUSTOMER BSFN - PortX (ms) |
|---------------------|-----------------------------------|-----------------------------------|
| Exec 5              | 693                               | 954                               |
| Exec 6              | 721                               | 865                               |
| Exec 7              | 676                               | 872                               |
| Exec 8              | 735                               | 851                               |
| Exec 9              | 691                               | 1100                              |
| Exec 10             | 779                               | 872                               |
| <b>Average (ms)</b> | <b>723.80</b>                     | <b>950.40</b>                     |

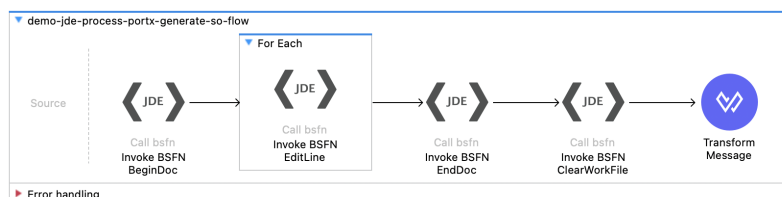
## 9. Scenario 07: Create Sales Order

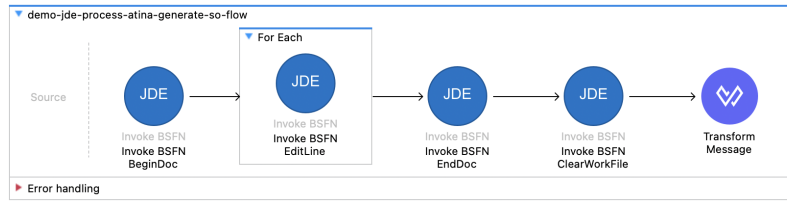
### 9.1. APIs used

- Web Service:
  - `oracle.e1.bssv.JP420000.SalesOrderManager.processSalesOrder`



- BSFNs (Transaction Processing):
  - `F4211FSBeginDoc`
  - `F4211FSEditLine` (per line)
  - `F4211FSEndDoc`
  - `F4211ClearWorkFile`





## 9.2. Cycle 1

| Execution           | Create Sales Order<br>WS (ms) | Create Sales Order<br>BSFN - Atina (ms) | Create Sales Order<br>BSFN - PortX (ms) |
|---------------------|-------------------------------|-----------------------------------------|-----------------------------------------|
| Exec 1              | 268003                        | 12223                                   | 135443                                  |
| Exec 2              | 87591                         | 2625                                    | 2602                                    |
| Exec 3              | 33132                         | 2674                                    | 2495                                    |
| <b>Average (ms)</b> | <b>129575.30</b>              | <b>5840.70</b>                          | <b>46846.70</b>                         |

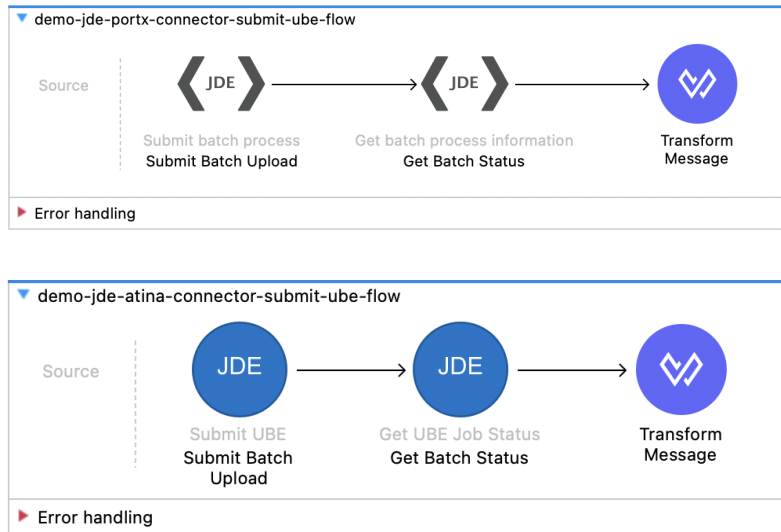
## 9.3. Cycle 2

| Execution           | Create Sales Order<br>WS (ms) | Create Sales Order<br>BSFN - Atina (ms) | Create Sales Order<br>BSFN - PortX (ms) |
|---------------------|-------------------------------|-----------------------------------------|-----------------------------------------|
| Exec 1              | 44898                         | 2582                                    | Time Out                                |
| Exec 2              | 31803                         | 6168                                    | 3216                                    |
| Exec 3              | 41434                         | 2525                                    | 8953                                    |
| Exec 4              | 35334                         | 2621                                    | 9438                                    |
| Exec 5              | 31830                         | 2498                                    | 2477                                    |
| Exec 6              | 32281                         | 2605                                    | 2496                                    |
| Exec 7              | 40250                         | 2511                                    | 2452                                    |
| Exec 8              | 34137                         | 2449                                    | 2696                                    |
| Exec 9              | 36355                         | 2725                                    | 9384                                    |
| Exec 10             | 31880                         | 2525                                    | 2485                                    |
| <b>Average (ms)</b> | <b>36020.20</b>               | <b>2920.90</b>                          | <b>4844.10</b>                          |

**NOTE** | Averages are calculated excluding the execution marked as Time Out.

# 10. Scenario 08: Submit UBE and retrieve status

This process submits a UBE and retrieves its status. In these runs, PortX is slightly faster than Atina.



## 10.1. Cycle 1

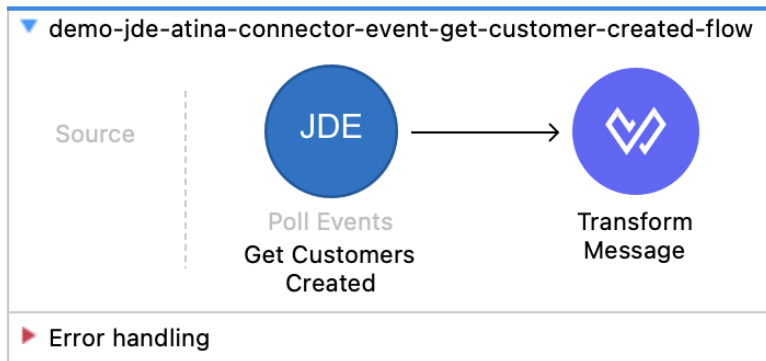
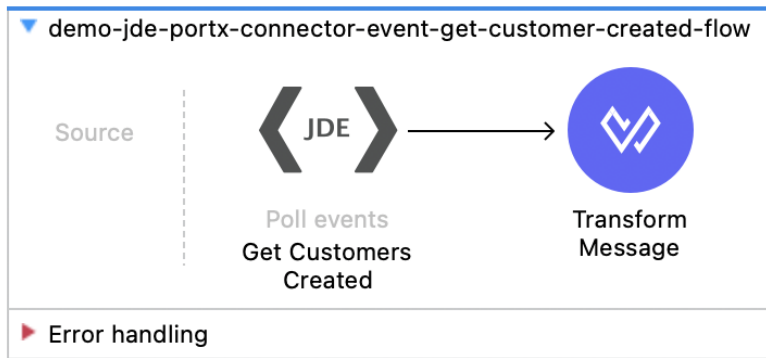
| Execution           | Submit UBE - Atina (ms) | Submit UBE - PortX (ms) |
|---------------------|-------------------------|-------------------------|
| Exec 1              | 5559                    | 1843                    |
| Exec 2              | 2083                    | 1791                    |
| Exec 3              | 1993                    | 1783                    |
| <b>Average (ms)</b> | <b>3211.70</b>          | <b>1805.70</b>          |

## 10.2. Cycle 2

| Execution           | Submit UBE - Atina (ms) | Submit UBE - PortX (ms) |
|---------------------|-------------------------|-------------------------|
| Exec 1              | 6441                    | 1791                    |
| Exec 2              | 2010                    | 1804                    |
| Exec 3              | 1972                    | 1732                    |
| Exec 4              | 1917                    | 1819                    |
| Exec 5              | 2029                    | 1818                    |
| Exec 6              | 1934                    | 1785                    |
| Exec 7              | 2077                    | 1838                    |
| Exec 8              | 1871                    | 1718                    |
| Exec 9              | 1971                    | 1808                    |
| Exec 10             | 2099                    | 1784                    |
| <b>Average (ms)</b> | <b>2432.10</b>          | <b>1789.90</b>          |

## 11. Scenario 09: Get Events (MBF Events)

This scenario retrieves customer-created events (2 records per request in these tests).



## 11.1. Cycle 1

| Execution           | Get Events - Atina (ms) | Get Events - PortX (ms) |
|---------------------|-------------------------|-------------------------|
| Exec 1              | 9827                    | Time Out                |
| Exec 2              | 3116                    | 4717                    |
| Exec 3              | 3152                    | 2946                    |
| <b>Average (ms)</b> | <b>5365.00</b>          | <b>3831.50</b>          |

**NOTE** | Averages are calculated excluding the execution marked as Time Out.

## 11.2. Cycle 2

| Execution | Get Events - Atina (ms) | Get Events - PortX (ms) |
|-----------|-------------------------|-------------------------|
| Exec 1    | 3079                    | Time Out                |
| Exec 2    | 3230                    | 4000                    |
| Exec 3    | 3146                    | 2951                    |
| Exec 4    | 3036                    | 3051                    |
| Exec 5    | 3135                    | 3050                    |
| Exec 6    | 3166                    | 3016                    |
| Exec 7    | 3142                    | 3054                    |
| Exec 8    | 3055                    | 3011                    |

| Execution           | Get Events - Atina (ms) | Get Events - PortX (ms) |
|---------------------|-------------------------|-------------------------|
| Exec 9              | 3068                    | 2866                    |
| Exec 10             | 3046                    | 2946                    |
| <b>Average (ms)</b> | <b>3110.30</b>          | <b>3105.00</b>          |

**NOTE** | Averages are calculated excluding the execution marked as Time Out.

## 12. Summary: Which connector performs better?

The table below summarizes steady-state performance (Cycle 2 where available). Lower is better. Averages are computed excluding executions marked as Time Out.

| Scenario                            | Atina Avg (ms) | PortX Avg (ms) | Winner | Speedup | Notes                                                         |
|-------------------------------------|----------------|----------------|--------|---------|---------------------------------------------------------------|
| Get Address Book (MBF – Inquire)    | 449.2          | 1046.3         | Atina  | 2.33x   | Cycle 2                                                       |
| Create Address Book (MBF – Add)     | 501.2          | 1527.9         | Atina  | 3.05x   | Cycle 2; PortX run includes a couple of ~5s outlier latencies |
| Delete Address Book (MBF – Delete)  | 608.3          | 792.5          | Atina  | 1.30x   | Cycle 2 (using the Atina average without the extreme outlier) |
| Get Customer (MBF + AB – Inquire)   | 852.3          | 2017.9         | Atina  | 2.37x   | Cycle 2                                                       |
| Create Customer (MBF + AB – Add)    | 2062.3         | 3975.9         | Atina  | 1.93x   | Cycle 2                                                       |
| Delete Customer (MBF + AB – Delete) | 723.8          | 950.4          | Atina  | 1.31x   | One cycle available                                           |



| Scenario                                         | Atina Avg (ms) | PortX Avg (ms) | Winner | Speedup | Notes                                                                                          |
|--------------------------------------------------|----------------|----------------|--------|---------|------------------------------------------------------------------------------------------------|
| Create Sales Order (BSFN Transaction Processing) | 2920.9         | 4844.1         | Atina  | 1.66x   | Cycle 2; PortX Exec 1 timed out and was excluded                                               |
| Submit UBE                                       | 2432.1         | 1789.7         | PortX  | 1.36x   | Cycle 2                                                                                        |
| Get Events                                       | 3110.3         | 3105.0         | Tie    | ~1.00x  | Cycle 2; PortX Exec 1 timed out and was excluded; remaining averages are essentially identical |

| Average (ms) | 7471.9 | 501.2 | 1527.9 Overall outcome: Atina is faster in most operations tested; PortX is faster in Submit UBE; Get Events is effectively a tie.